Impact of magnetic properties on the Casimir torque between anisotropic metamaterial plates

Gang Deng, Zhong-Zhu Liu, and Jun Luo*
Department of Physics, Huazhong University of Science and Technology, Wuhan 430074, People's Republic of China
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The quantized surface mode technique is used to calculate the Casimir torque between two parallel anisotropic metamaterial plates with in-plane optical axes, and our main concern is focused on the impact of the magnetic properties of the plates on the Casimir torque. Our result shows that at small separation, the Casimir torque between the two plates with frequency dependent permeabilities is larger than that between two nonmagnetic plates, while at large separation it is smaller. This can be explained as a result of the impact of both magnetic properties and material dispersion of the plates. The impact of the Drude background in connected metallic metamaterial is also discussed. These phenomena provide us with new understanding about the Casimir effect and show great potential in application.

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I. INTRODUCTION

The Casimir effect caused by the quantum fluctuations of electromagnetic fields is one of the most remarkable predictions of quantum electrodynamics [1–5]. Great effort has been put into both theoretical and experimental studies on the Casimir effect [4,5], as it plays an important role in various fields of physics [4].

As we know, Lifshitz's original theory for the Casimir force is only applicable to isotropic dielectrics [2,3]. However, if the plates are birefringent, the zero-point energy between the plates will be orientation dependent [6], which will result in a Casimir torque. Most researches on the Casimir torque are based on the assumption that the two plates are nonmagnetic [6–12]. In a previous work of our group [13], the nontrivial magnetic permeabilities of the plates have been considered in the Casimir torque calculation for the first time. However, in that work [13] the permittivities and permeabilities were assumed to be constant rather than frequency dependent, and some interesting effects, such as attractive-repulsive transition of the Casimir force, would not occur as mentioned in our recent work [14].

It is notable that the nontrivial magnetic permeability might lead to a repulsive Casimir force [13–20]. Especially, the attractive-repulsive transition of the Casimir force will happen if material dispersion is also taken into account at the same time [14]. Consequently, a question naturally arises: what new Casimir torque phenomena can be brought out by considering both the nontrivial permeabilities and material dispersion of the plates. Unfortunately, no clear clarification to this has been reported yet. On the other hand, as the metamaterials usually have special electric and magnetic responses [21], considerable attention has been paid to the Casimir effect between metamaterial bodies in recent years [14–17]. From the standpoint of application, the Casimir torque may be expected to be used in rotation control in micromachines [4,5]. Thus, it is necessary to study the possible new features of the Casimir torque between anisotropic metamaterial plates with frequency dependent permittivities and permeabilities. In the present work we still use the quantized surface mode technique [10,13,14,22] to calculate the Casimir torque between two anisotropic metamaterial plates, and the frequency dispersion of the permittivities and permeabilities of the plates has also been taken into account (the permittivities and permeabilities were assumed to be constant in our previous work [13]). In Ref. [14], there is no Casimir toque, as we have assumed the optical axes to be parallel. In the present work we give up this assumption, and the angle between the two optical axes is arbitrary. Our major concern is focused on the contribution of the magnetic property (frequency dependent permeability) to the Casimir torque. The result shows that at small separation the magnetic property gives a positive contribution to the Casimir torque, while at large separation, it gives a negative contribution to the Casimir torque. This can be explained as a result of the combined impact of the magnetic properties and material dispersion of the plates. This effect not only provides us with important new information about the Casimir effect but also shows great potential in rotation control in micromechanical application. We will discuss this phenomenon and relevant effects in detail in the following sections, and this work is also an extension of our previous works [10,13,14].

II. EVALUATION OF THE CASIMIR TORQUE

The system considered is shown in Fig. 1. Two uniaxial plates (with the diameter D and the thickness d) made of different materials are kept parallel to each other and separated with a distance a in vacuum. The x-y plane is chosen to be parallel to the surfaces of the plates, and the optic axes of the two plates are in the x-y plane. θ is the angle between the optical axis of plate 1 and x direction, while φ is the angle between the optic axes of the two plates. Assuming $D \gg a$ and $d \gg a$, it is reasonable to disregard the edge effect and finite thickness effect. And the space can be approximately considered to be divided into three regions with corresponding permittivity tensors ε_1, ε_0, ε_2 and permeability tensors μ_1, μ_0, μ_2, as shown in Fig. 2.

According to the quantized surface mode technique, we only need to consider the zero-point energy associated with

*junluo@mail.hust.edu.cn

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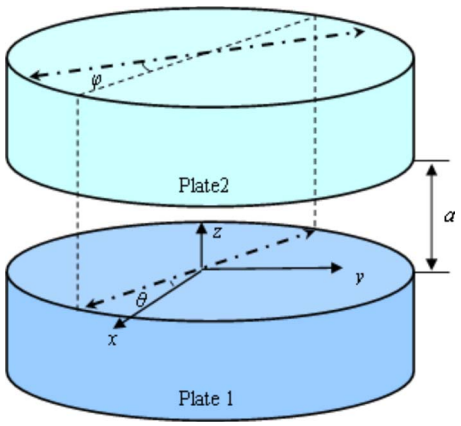


FIG. 1. (Color online) Two uniaxial plates with in-plane optic axes are separated by vacuum with a distance a. The surfaces of the plates are parallel to the x-y plane. And θ is the angle between the optic axis of plate 1 and x direction, while φ is the angle between the optic axes of the two plates.

the surface modes q which are exponentially decaying when $z>a$ and $z<0$ [10,13,14,22]. Because each region in Fig. 2 is considered as homogenous, the electric and magnetic fields can be expressed as [10,13,14,22]

$$\vec{E}_q = iNe^{ik_x x+ik_y y}(a_q\vec{e}_q e^{ik_z z} - a_q^\dagger \vec{e}_q^* e^{ik_z z}), \tag{1}$$

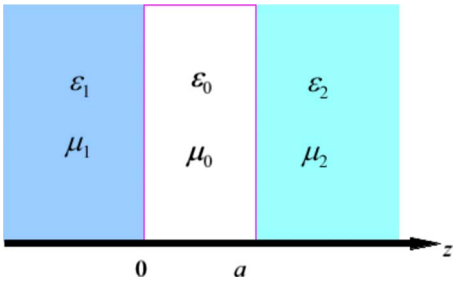


FIG. 2. (Color online) The schematic configuration of vacuum space within two uniaxial plates. The optical property of each region is described by corresponding permittivity and permeability.

$$\vec{H}_q = Ne^{ik_x x+ik_y y}(a_q\vec{h}_q e^{ik_z z} + a_q^\dagger \vec{h}_q^* e^{ik_z z}), \tag{2}$$

where N is the normalized factor and $\vec{k}_\parallel=(k_x,k_y)$ and $\vec{k}_z$ are the propagation vectors in the x-y plane and in the z direction. The parameters a_q and $a_q^\dagger$ are the usual creation and annihilation operators, respectively. The parameters $\vec{e}_q$ and $\vec{h}_q$ are the electromagnetic field polarization vectors and ω is the angular frequency.

We can choose $\vec{k}_\parallel$ to be parallel to the x direction and then the wave vector can be written as $\vec{k}=(k_x,k_y,k_z)=K_0(\alpha,0,\gamma)$, with $K_0=\omega c^{-1}$, $K_\parallel=K_0\alpha$, and $K_\perp=K_0\gamma$. For the case that the two plates are dielectrically anisotropic but magnetically isotropic, the permittivity and permeability tensors of the two plates can be expressed as

$$\hat{\varepsilon}_1 = \begin{pmatrix} \varepsilon_{1,\parallel}\cos^2\theta + \varepsilon_{1,\perp}\sin^2\theta & (\varepsilon_{1,\parallel}-\varepsilon_{1,\perp})\cos\theta\sin\theta & 0 \\ (\varepsilon_{1,\parallel}-\varepsilon_{1,\perp})\cos\theta\sin\theta & \varepsilon_{1,\parallel}\sin^2\theta + \varepsilon_{1,\perp}\cos^2\theta & 0 \\ 0 & 0 & \varepsilon_{1,\perp} \end{pmatrix}, \tag{3}$$

$$\hat{\varepsilon}_2 = \begin{pmatrix} \varepsilon_{2,\parallel}\cos^2(\theta+\varphi) + \varepsilon_{2,\perp}\sin^2(\theta+\varphi) & (\varepsilon_{2,\parallel}-\varepsilon_{2,\perp})\cos(\theta+\varphi)\sin(\theta+\varphi) & 0 \\ (\varepsilon_{2,\parallel}-\varepsilon_{2,\perp})\cos(\theta+\varphi)\sin(\theta+\varphi) & \varepsilon_{2,\parallel}\sin^2(\theta+\varphi) + \varepsilon_{2,\perp}\cos^2(\theta+\varphi) & 0 \\ 0 & 0 & \varepsilon_{2,\perp} \end{pmatrix}, \tag{4}$$

$$\hat{\mu}_1 = \begin{pmatrix} \mu_1 & 0 & 0 \\ 0 & \mu_1 & 0 \\ 0 & 0 & \mu_1 \end{pmatrix}, \tag{5}$$

$$\hat{\mu}_2 = \begin{pmatrix} \mu_2 & 0 & 0 \\ 0 & \mu_2 & 0 \\ 0 & 0 & \mu_2 \end{pmatrix}, \tag{6}$$

where the subscripts $\parallel$ and $\perp$ indicate the tensor components parallel and perpendicular to the optical axes, respectively. A variable δ_i, defined as $\varepsilon_{i,\parallel}=(1+\delta_i)\varepsilon_{i,\perp}$, is introduced to describe the degree of anisotropy of the plate i. For convenience, we will write $\varepsilon_{i,\perp}=\varepsilon_i$ in following discussions. Hence, for small δ_1 and δ_2, the Casimir energy per unit area between the plates at zero temperature can be expressed as ($\xi=-i\omega$ has been introduced [2,3])

$$E(a,\varphi) = \frac{\hbar}{4\pi^2c^2}\int_1^\infty pdp\int_0^\infty \xi^2 d\xi \times [E_0 + E_1\delta_1 + E_2\delta_2 + E_3\delta_1\delta_2], \tag{7}$$

where c is the speed of light in vacuum and $\hbar$ is the Plank constant divided by 2π. The variable p is introduced as $p^2=1-\alpha^2$. E_0, E_1, E_2, and E_3 in Eq. (7) are expressed as

$$E_0(a) = \ln\Delta_{\mathrm{TE}} + \ln\Delta_{\mathrm{TM}}, \tag{8}$$

$$E_1(a) = \frac{\varepsilon_1\mu_1 m_1(1-m_2)}{2t_1^2(1+m_1)^2(1+m_2)}\Delta_{\mathrm{TE}}^{-1} - \frac{w_1(1-w_2)}{2(1+w_1)^2(1+w_2)}\Delta_{\mathrm{TM}}^{-1}, \tag{9}$$

$$E_2(a) = \frac{\varepsilon_2\mu_2 m_2(1-m_1)}{2t_2^2(1+m_2)^2(1+m_1)}\Delta_{\mathrm{TE}}^{-1} - \frac{w_2(1-w_1)}{2(1+w_2)^2(1+w_1)}\Delta_{\mathrm{TM}}^{-1}, \tag{10}$$

$$E_3(a) = \left[\frac{m_1 m_2 \Delta_{\mathrm{TM}}^{-1}\Delta_{\mathrm{TE}}^{-1}}{4(1+m_1)(1+m_2)(1+w_1)(1+w_2)} - \frac{\varepsilon_1\mu_1\varepsilon_2\mu_2}{8t_1^2t_2^2}\frac{m_1 m_2 \Delta_{\mathrm{TE}}^{-2}}{(1+m_1)^2(1+m_2)^2} - \frac{w_1 w_2 \Delta_{\mathrm{TM}}^{-2}}{8(1+w_1)^2(1+w_2)^2}\right] e^{-2\xi pa/c}\cos 2\varphi + C, \tag{11}$$

where $\Delta_{\mathrm{TE}} = 1-[(1+m_1)(1+m_2)/(1-m_1)(1-m_2)]e^{2\xi pa/c}$, $\Delta_{\mathrm{TM}} = 1-[(1+w_1)(1+w_2)/(1-w_1)(1-w_2)]e^{2\xi pa/c}$, $t_i^2 = -p^2 - \varepsilon_i\mu_i + 1$, $m_i = (\mu_i p)/\sqrt{p^2+\varepsilon_i\mu_i-1}$, $w_i = \sqrt{p^2+\varepsilon_i\mu_i-1}/(\varepsilon_i p)$ $(i=1,2)$, and C is a φ independent term.

One can see that the first three terms in Eq. (7) do not contribute to the torque as they are φ independent. Only the term $E_3\delta_1\delta_2$ is effective to the toque. This in some sense shows that the Casimir torque is a phenomenon which reflects the relation between the anisotropies of the two plates. Thus the Casimir energy in Eq. (7) can be written in the following form:

$$E(a,\varphi) = E(a) - \frac{1}{2}T(a)\cos 2\varphi. \tag{12}$$

The corresponding Casimir torque is

$$\tau(a,\varphi) = -\frac{\partial E(a,\varphi)}{\partial\varphi} = -T(a)\sin 2\varphi, \tag{13}$$

where $T(a)$ is the amplitude of the torque which can be expressed as

$$T(a) = \frac{\hbar\delta_1\delta_2}{2\pi^2c^2}\int_1^\infty pdp\int_0^\infty \xi^2 d\xi e^{-2\xi pa/c} \times \left[\frac{m_1 m_2 \Delta_{\mathrm{TM}}^{-1}\Delta_{\mathrm{TE}}^{-1}}{4t_0^2(1+m_1)(1+m_2)(1+w_1)(1+w_2)} - \frac{\varepsilon_1\mu_1\varepsilon_2\mu_2}{8t_1^2t_2^2}\frac{m_1 m_2 \Delta_{\mathrm{TE}}^{-2}}{(1+m_1)^2(1+m_2)^2} - \frac{w_1 w_2 \Delta_{\mathrm{TM}}^{-2}}{8(1+w_1)^2(1+w_2)^2}\right]. \tag{14}$$

The dispersion behavior of permittivity and permeability of materials usually can be described by Drude-Lorentz model [23],

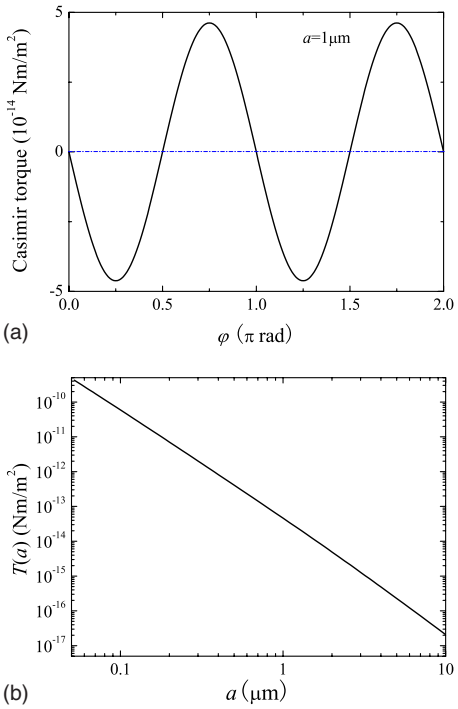


FIG. 3. (Color online) (a) The Casimir torque between two uniaxial metamaterial plates as a function of φ, the angle between the optic axes of the two plates. Parameters are as follows: $\Omega = 1.47\times10^{16}$ rad/s is the plasma frequency of silver [24]; for plate 1, $\Omega_{D1}=\Omega$, $\gamma_{D1}=\gamma_{e1}=0.005\Omega$, $\Omega_{e1}=0.04\Omega$, $\omega_{e1}=0.1\Omega$, $f_1=0.001$, $\delta_1=0.1$, and $\mu_1=1$; for plate 2, $\Omega_{D2}=\Omega$, $\gamma_{D2}=\gamma_{e2}=\gamma_{m2}=0.005\Omega$, $\Omega_{e2}=0.04\Omega$, $\Omega_{m2}=0.3\Omega$, $\omega_{e2}=\omega_{m2}=0.1\Omega$, $f_2=0$, and $\delta_2=0.1$. (b) The amplitude of the torque $T(a)$ as a function of the separation a.

$$\varepsilon(i\xi) = 1 + \frac{\Omega_{pe}^2}{\omega_e^2+\xi^2+\gamma_e\xi} \tag{15}$$

and

$$\mu(i\xi) = 1 + \frac{\Omega_{pm}^2}{\omega_m^2+\xi^2+\gamma_m\xi}. \tag{16}$$

In these representations, Ω_{pe} (Ω_{pm}) is the plasma frequency or electric (magnetic) oscillating strength, ω_e (ω_m) is the electric (magnetic) resonance frequency, and γ_e (γ_m) is the electric (magnetic) dissipation factor. For connected metallic metamaterial, the dielectric function is usually considered to have a small Drude background in addition to the usual Drude-Lorentz response, and the permittivity can be modeled as [14–17]

$$\varepsilon(i\xi) = 1 + (1-f)\frac{\Omega_e^2}{\omega_e^2+\xi^2+\gamma_e\xi} + f\frac{\Omega_D^2}{\xi^2+\gamma_D\xi}, \tag{17}$$

where Ω_D and γ_D are the plasma frequency and the dissipation factor of the metallic structure in the metamaterial. The fill factor f roughly represents the fraction of this metallic structure in the metamaterial.

The Casimir torque between different anisotropic plates can be calculated using Eqs. (13)–(17). Figure 3(a) shows the Casimir torque between two metamaterial plates separated

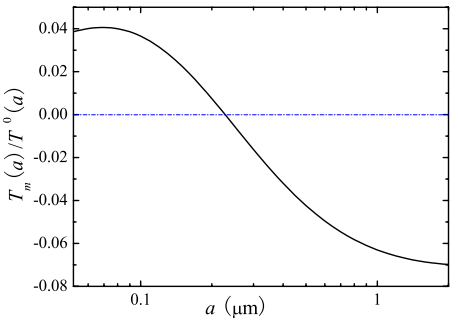


FIG. 4. (Color online) The magnetic impact on the Casimir torque at different separations is calculated with the same parameters used in Fig. 3. $T^0(a)$ is calculated with same parameters except for $\mu_1=\mu_2=1$.

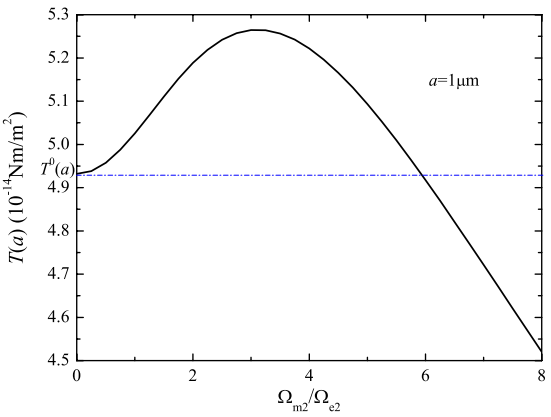


FIG. 5. (Color online) The Casimir torque $[T(a)]$ versus Ω_{m2}/Ω_{e2}. Parameters used here are the same as those used in Fig. 4 except Ω_{m2}.

by 1 μm. The torque varies with the angle between the two optical axes with periodicity equal to π and its maximum magnitude occurs at $\varphi=\pi/4$ and $\varphi=3\pi/4$. The angles $\varphi=\pi/4$ and $\varphi=\pi$ are the stable orientations since $E(a,\varphi)$ reaches its minimum value at these orientations according to Eqs. (12) and (14). This result consists of the conclusion given by Munday *et al.* [8]. The amplitude of the total Casimir torque $T(a)$ falls rapidly as the separation increases, as shown in Fig. 3(b).

III. IMPACT OF MAGNETIC PROPERTIES

As mentioned previously, $T(a)$ is the amplitude of the total Casimir torque between two plates with nontrivial magnetic permeabilities. Let $T^0(a)$ be the amplitude of Casimir torque between two nonmagnetic plates, and it is calculated with the same parameters in calculating $T(a)$ except for $\mu_1=\mu_2=1$. The difference $T_m(a)=T(a)-T^0(a)$ describes the contribution of magnetic property to the Casimir torque, and we can call $T_m(a)$ the magnetic Casimir torque. As shown in Fig. 4, $T_m(a)$ is positive at small separation, which means that the contribution of the magnetic property is to increase the Casimir torque and it helps to make the two optic axes parallel. However, at large separation, the contribution of the magnetic property $[T_m(a)]$ changes into negative and prevents the two optical axes to be parallel, which can be considered as a "repulsive" torque. This interesting phenomenon seems similar to the attractive-repulsive transition of the Casimir force in Ref. [14]. However, one should notice that the sign of the total Casimir torque $T(a)$ does not change. What changes into negative is the magnetic Casimir torque, $T_m(a)$. That is also why the word repulsive is marked with quotation marks.

To explain this interesting phenomenon, we calculate $T(a)$ with different values of Ω_{m2}/Ω_{e2}. As shown in Fig. 5, the Casimir torque increases with Ω_{m2}/Ω_{e2} when Ω_{m2}/Ω_{e2} is small. However, when Ω_{m2}/Ω_{e2} becomes larger the Casimir torque starts to decrease as Ω_{m2}/Ω_{e2} increases. And for very large Ω_{m2}/Ω_{e2} [25], the Casimir torque $T(a)$ is always smaller than $T^0(a)$. As we have set $\omega_{e2}=\omega_{m2}$, $\gamma_{e2}=\gamma_{m2}$, and $f_2=0$, the larger Ω_{m2}/Ω_{e2} always corresponds to the larger μ_2/ε_2. Thus, we can conclude that $T(a)<T^0(a)$ $[T_m(a)<0]$ can be satisfied only for large μ_2/ε_2. This can help us to explain the interesting behavior observed in Fig. 4. When the permittivities and permeabilities are frequency dependent, the plates have different responses to different frequency modes. The main contribution to the Casimir effect comes from the modes whose wavelengths are of the order of the distance between the two plates [2,3]. We can call these modes the active modes. For very small separation, the active modes are mainly high frequency modes. The magnetic responses of the plates to these modes are negligible, and the electric responses are dominant. Thus, μ_2/ε_2 is small, and $T_m(a)$ is positive. As the separation becomes larger, the low frequency modes, to which the plates have significant magnetic responses, become more important. This makes μ_2/ε_2 larger. Consequently, $T_m(a)$ changes into negative and exhibits like a repulsive torque. This phenomenon also shows the competition between the values of ε and μ.

Similar phenomenon happened in work of Munday *et al.*, in which the Casimir torque between two nonmagnetic birefringent plates was calculated [8]. They found that the torque in liquid ethanol was smaller than in vacuum at large distance, while it was larger than in vacuum at short distance. This phenomenon may also be explained as a result of the material dispersion. The only difference is that the competition occurred between the values of the permittivities of three materials rather than between the values of the permittivity and permeability of one plate in our case.

IV. IMPACT OF DRUDE BACKGROUND IN CONNECTED METALLIC METAMATERIAL

In connected metallic metamaterials the metallic part is interconnected, so there is a Drude-type conductivity contribution to the dielectric function as shown in Eq. (17) [16]. The fill factor f is used to quantify the fraction of the metallic structure in the metamaterial. Even a small f can significantly change the permittivity of the plate according to Eq. (17). Therefore, it is necessary to discuss the impact of the Drude background on the Casimir torque.

We calculate $T_m(a)/T^0(a)$ at different separations with different f_2, the fill factor of plate 2. The result is plotted in Fig. 6. All the other parameters are the same as that used in

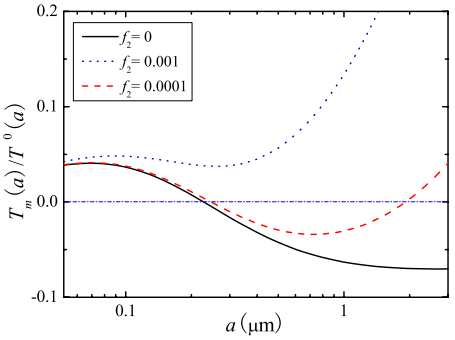


FIG. 6. (Color online) The magnetic impact on the Casimir torque at different separations with different f_2. All the other parameters are the same as that used in Fig. 4.

Fig. 4. From Fig. 6 we can find that $T_m(a)$ is negative only in an intermediate separation range and goes back to positive again at large separation. This can be explained as a result of the contribution of the Drude background. As the separation becomes even larger, the active modes between the plates are mainly very low frequency modes. Thus, ε_2 becomes very large due to the Drude background, while μ_2 does not change too much according to Eqs. (16) and (17), which makes μ_2/ε_2 small. Consequently, $T_m(a)$ again changes into positive. Notice that the magnetic Casimir torque is negative in a narrow intermediate distance range only for very small f_2. As shown in Fig. 6, for larger fill factor f_2 (f_2=0.001), $T_m(a)$ is always positive. This is because the large Drude background makes $\varepsilon_2 > \mu_2$ satisfied at all separations.

V. CONCLUSION

The Casimir torque between two anisotropic metamaterial plates has been calculated using quantized surface mode technique, and the frequency dependent permittivities and permeabilities have been considered. At small separation the magnetic property gives a positive contribution to the total Casimir torque $[T(a)]$. As the separation increases, the contribution of the magnetic property $[T_m(a)]$ changes to negative and prevents the two optical axes to be parallel, which seems like a repulsive torque. With nearly vanishing Drude background, $T_m(a)$ will go back to positive at even larger separation. All these can be explained as the combined impact of magnetic properties and material dispersion of the plates. For present technology, performing an accurate experiment to observe the impact of magnetic property on the Casimir effect is still difficult since present metamaterials can only have a narrow band where $\mu > 1$ in optical frequency range and it is still difficult to achieve large values of μ/ε [5,21,25]. Nevertheless, it does show us some interesting new phenomena which, if successfully observed, would provide us some new understanding about the Casimir effect and electromagnetic theory for real media. From the viewpoint of application, these may be expected to be used in rotation control in micromechanical systems in the future.

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